

U.S. Flight Delay Prediction and Airport Risk Segmentation: A Reproducible Data Mining Study

Yiting Li*

Sihan Yan*

Junyang Wang*

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Abstract

This project studies two linked questions using 2019 U.S. domestic flight records derived from the Bureau of Transportation Statistics and distributed through the Kaggle Flight Delay dataset. First, can a flight’s risk of arriving at least 15 minutes late be predicted using only information known before departure? Second, do U.S. airports fall into meaningful delay-risk segments when they are summarized by operating profiles rather than individual flights? To answer the prediction question, the project constructs a cleaned 10% stratified sample and compares an always-on-time baseline, LASSO logistic regression, random forest, and extra-trees classifier on a common train/validation/test split. To answer the segmentation question, it builds airport-level operating profiles and applies k -means clustering. The random forest has the strongest predictive performance, with a holdout ROC AUC of 0.671, PR AUC of 0.343, and top-decile delay lift of 2.21. The clustering analysis identifies three interpretable airport types: Congested Hubs, Small High-Volatility Airports, and Lower-Risk Diversified Airports. The main conclusion is that useful pre-departure delay-risk information exists, but predictive performance remains moderate because important drivers such as weather, air traffic control disruptions, maintenance issues, and network spillovers are not directly observed.

1 Introduction and Motivation

Flight delays are economically and operationally important. Travelers care because delays increase missed connections, travel time uncertainty, and personal inconvenience. Airlines care because delays disrupt crew scheduling, aircraft rotations, gate utilization, and customer satisfaction. Airports and regulators care because recurring congestion and delay concentration can reveal deeper operational bottlenecks. A data mining project on flight delays is therefore useful only if it goes beyond technical model fitting and instead answers a practical question about delay risk before departure and about structural differences across airports.

This project addresses that practical need in two complementary ways. The first task is predictive: estimate whether a given flight will arrive at least 15 minutes late using only information that could plausibly be known before the aircraft departs. The second task is descriptive and unsupervised: group airports according to operating patterns and delay profiles to understand whether some airports systematically face different delay environments. The predictive analysis is useful for advance risk assessment, while the clustering analysis provides system-level context for interpreting where elevated delay risk is concentrated.

The core contribution is not a claim of perfect prediction. Instead, the contribution is a transparent and reproducible workflow that combines motivated feature design, interpretable exploratory analysis, fair model comparison, and a clustering summary that helps explain why some parts of the flight system appear structurally riskier than others.

*All authors contributed equally to this work.

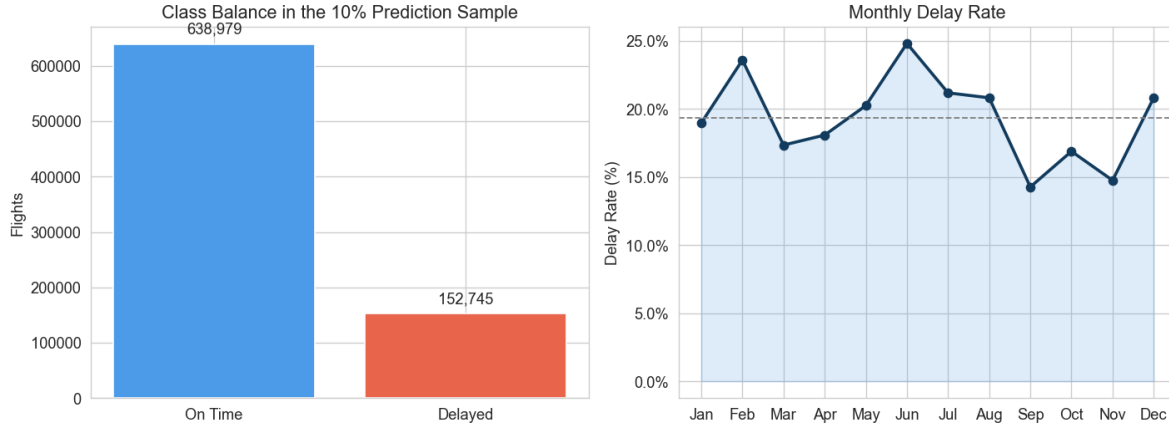


Figure 1. Class balance and monthly delay trend in the cleaned 2019 sample. The delayed class is the minority class, and delay risk varies substantially by month.

2 Data Description

The raw dataset is the 2019 U.S. domestic flight delay data distributed through Kaggle and derived from the BTS Marketing Carrier On-Time Performance database. The repository expects the raw parquet file at `data/Combined_Flights_2019.parquet`. The project also generates two cleaned derivative datasets: `data/flights_2019_sample.parquet` for the prediction workflow and `data/airport_profiles_2019.csv` for airport clustering.

The prediction target is `ArrDel15`, an indicator equal to one when a flight arrived at least 15 minutes late. The analysis intentionally restricts predictors to variables available before departure. These include carrier, origin airport, destination airport, month, day-of-month timing features, scheduled departure block, scheduled elapsed time, and route distance. Excluding post-departure variables matters because the substantive question is pre-departure risk assessment rather than after-the-fact explanation.

Data preparation produces a cleaned 10% prediction sample stratified jointly by `Month` and `ArrDel15`. This design helps preserve both seasonal structure and class balance. The final cleaned sample contains 791,724 flights with a delay rate of 19.29%. For modeling, the pipeline uses a capped 200,000-row modeling subset for computational feasibility while preserving the same feature engineering and split logic. For clustering, the project aggregates flights into airport-level operating profiles using delay behavior, traffic volume, route structure, carrier diversity, destination diversity, evening share, and related summary measures. To avoid unstable micro-clusters dominated by extremely small airports, the final clustering step retains airports with at least 1,000 annual flights.

The dataset has important limitations. It covers one calendar year, so broader temporal generalization should be treated cautiously. It also lacks direct weather measurements, air traffic control constraints, maintenance shocks, and propagation of upstream delays across aircraft rotations. Those omissions matter when interpreting why predictive performance is meaningful but far from perfect.

3 Exploratory Data Analysis

The exploratory phase serves two purposes: it helps the reader understand the structure of delay risk and motivates later modeling choices. Overall, the cleaned sample is imbalanced but not an extreme rare-event problem: 19.29% of sampled flights are delayed. This immediately makes raw accuracy an incomplete evaluation metric, because a naive always-on-time rule can appear strong despite being uninformative for ranking delay risk.

Several descriptive patterns stand out. First, delay risk is seasonal: June is the highest-delay month and September is the lowest-delay month in the current run. Second, scheduled departure timing matters.

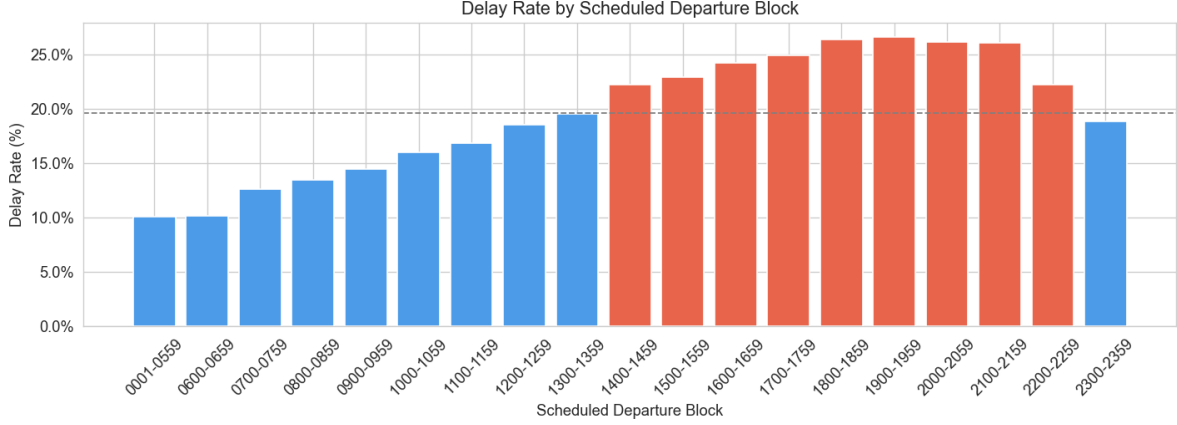


Figure 2. Delay rate by scheduled departure time block. Delay risk rises through the day and peaks in the evening.

Late-evening flights are materially riskier than early-morning flights, with the 1900–1959 block emerging as the highest-risk scheduled departure window and the 0001–0559 block as the lowest-risk block.

Third, airline and airport heterogeneity is substantial. Congested airports such as EWR, ORD, and LGA appear repeatedly among the highest-risk locations, suggesting that route and airport context should matter in the predictive task.

These descriptive results justify the later analytic choices. A sparse linear classifier is useful because some risk signals, such as specific airports or time blocks, should be interpretable directly. A nonlinear model is also justified because interactions among timing, route characteristics, and airports may be hard to capture with a single linear decision boundary. Finally, an unsupervised clustering stage is motivated by visible airport heterogeneity in volume, volatility, and delay exposure.

4 Methods

4.1 Data Preparation and Sampling

The data preparation workflow cleans the raw flight records, retains predictors known before departure, and constructs the binary target for delay classification. The cleaned data are sampled at 10% using strata defined by month and delay status so that both seasonal and class structure remain represented. This sampling strategy is preferable to an arbitrary simple random sample because it reduces the risk of underrepresenting important month-specific delay patterns.

4.2 Predictive Modeling

The predictive task compares four models:

1. an always-on-time baseline,
2. LASSO logistic regression,
3. tuned random forest, and
4. extra-trees classifier as an additional nonlinear robustness benchmark.

The baseline clarifies how misleading raw accuracy can be under class imbalance. LASSO logistic regression is appropriate because it offers a directly interpretable linear classifier with coefficient shrinkage and implicit variable selection. This matters because substantive interpretation is one of the goals of the project, and LASSO coefficients can identify airports, route structures, and departure windows associated with higher or lower delay risk.

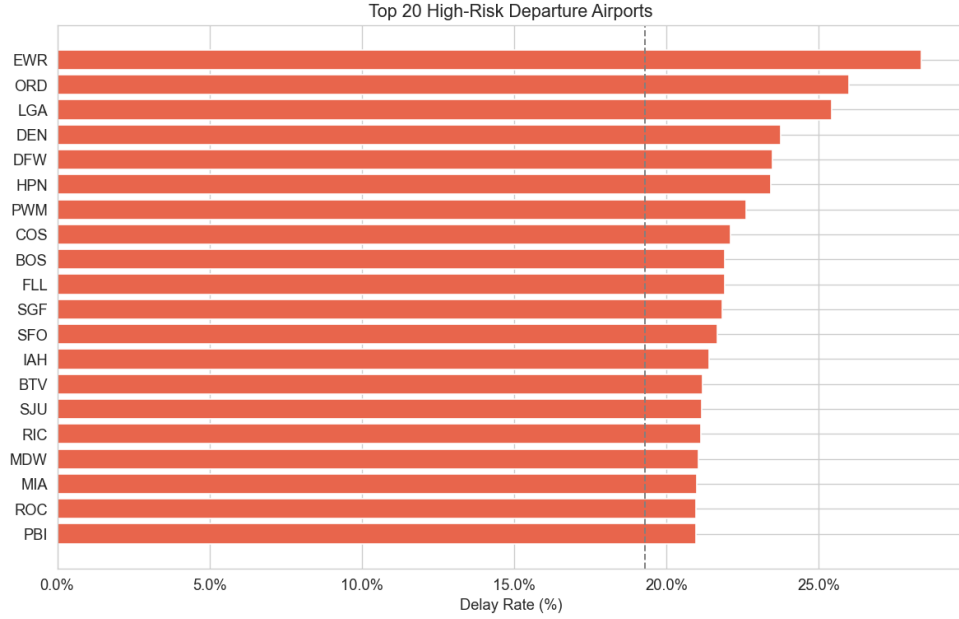


Figure 3. Highest-risk origin airports in the cleaned sample. High-delay airports include both major congested hubs and smaller volatile airports.

The random forest and extra-trees models are included because the delay process likely contains nonlinearities and interactions. Tree ensembles can capture threshold effects and conditional relationships that a linear model may miss. To make the comparison fair, the predictive models use the same train/validation/test split. Hyperparameter tuning for the ensemble models is performed on a tuning subset, and the final evaluation is reported on a held-out test set.

4.3 Evaluation Strategy

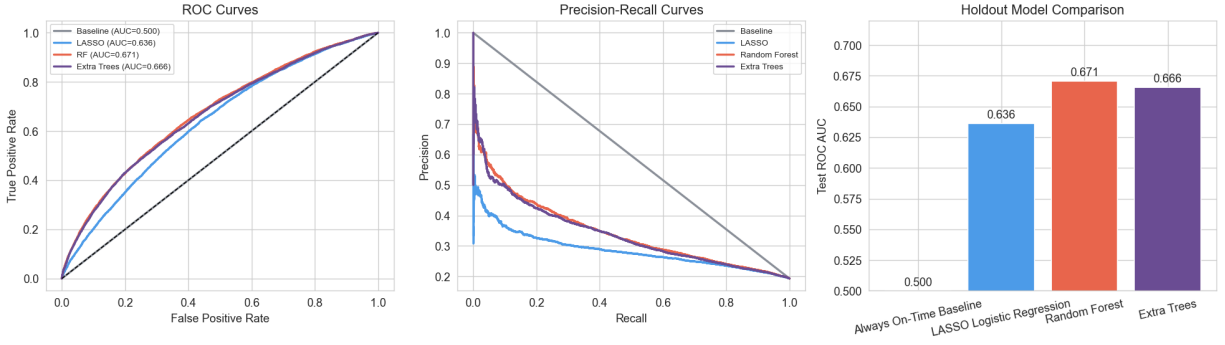
Model evaluation uses ROC AUC, PR AUC, balanced accuracy, delayed-class precision and recall, Brier score, and top-decile lift. ROC AUC is useful as a general ranking measure, but PR AUC is especially relevant because the delayed class is a minority. Balanced accuracy is included so that sensitivity and specificity are not collapsed into a misleading overall accuracy figure. Top-decile lift is included because an operational user might care whether the model’s highest-risk flights are meaningfully enriched in true delays relative to the base rate. Calibration diagnostics are also saved because practical deployment often depends on probability quality and threshold choices, not just on AUC.

4.4 Airport Segmentation

For the clustering task, the project aggregates airport-level features such as delay rate, average arrival delay, upper-tail delay severity, distance patterns, carrier diversity, destination diversity, evening traffic share, and flights per day. k -means clustering is then applied to the airport profiles. Diagnostic summaries are computed for $k = 2$ through $k = 6$. Although $k = 2$ achieves the highest silhouette score, the final segmentation uses $k = 3$ because it provides a more useful and interpretable typology for discussion. That choice is defensible because clustering is not only an optimization exercise; it is also an interpretive tool.

Table 1. Holdout predictive performance by model.

Model	ROC AUC	PR AUC	Accuracy	Balanced Acc.	Precision	Recall	Top-Decile Rate	Lift
Always On-Time Baseline	0.500	0.193	0.807	0.500	0.000	0.000	19.3%	1.00
LASSO Logistic Regression	0.636	0.286	0.587	0.599	0.260	0.618	33.6%	1.74
Random Forest	0.671	0.343	0.671	0.622	0.303	0.543	42.7%	2.21
Extra Trees	0.666	0.338	0.620	0.616	0.279	0.610	41.8%	2.17

**Figure 4.** ROC, precision-recall, and headline AUC comparison. The tree-based models outperform the baseline and LASSO model, especially in PR AUC and top-risk concentration.

5 Results

5.1 Predictive Performance

Table 1 summarizes the final held-out model performance. The always-on-time baseline achieves accuracy of 0.807, but this result is not impressive because it simply mirrors the fact that only about 19.3% of flights are delayed. Its ROC AUC is 0.500 and PR AUC is 0.193, confirming that it has no ranking power.

The LASSO logistic regression yields a cross-validated training AUC of 0.637, a holdout ROC AUC of 0.636, a holdout PR AUC of 0.286, and a balanced accuracy of 0.599. These values indicate that a relatively simple, interpretable model can recover meaningful signal from pre-departure variables.

The random forest performs best overall, with cross-validated tuning AUC of 0.654, holdout ROC AUC of 0.671, holdout PR AUC of 0.343, and balanced accuracy of 0.622. Its top-decile delay lift is 2.21, compared with 1.74 for the LASSO model. The extra-trees model also performs competitively, with holdout ROC AUC of 0.666 and top-decile lift of 2.17, which strengthens confidence that the nonlinear signal is real rather than model-specific. In practical terms, the random forest’s highest-risk decile contains about 42.7% delayed flights versus 19.3% in the full sample. That is a meaningful increase in concentration even though the model is far from deterministic.

5.2 Interpretable Signals

The LASSO coefficients highlight airports such as EWR, LGA, and SFO on the positive side, while low-risk locations and overnight departure windows appear on the negative side. This result aligns with the EDA evidence that delay risk is concentrated at specific congested locations and varies across time windows.

The random forest feature-group importance results point to departure hour, day-of-month timing, distance, scheduled elapsed time, and month as especially influential. Table 2 reports the top feature groups. Together, the LASSO and random forest tell a consistent story: both calendar timing and airport context matter, and delay risk is not evenly distributed across the network.

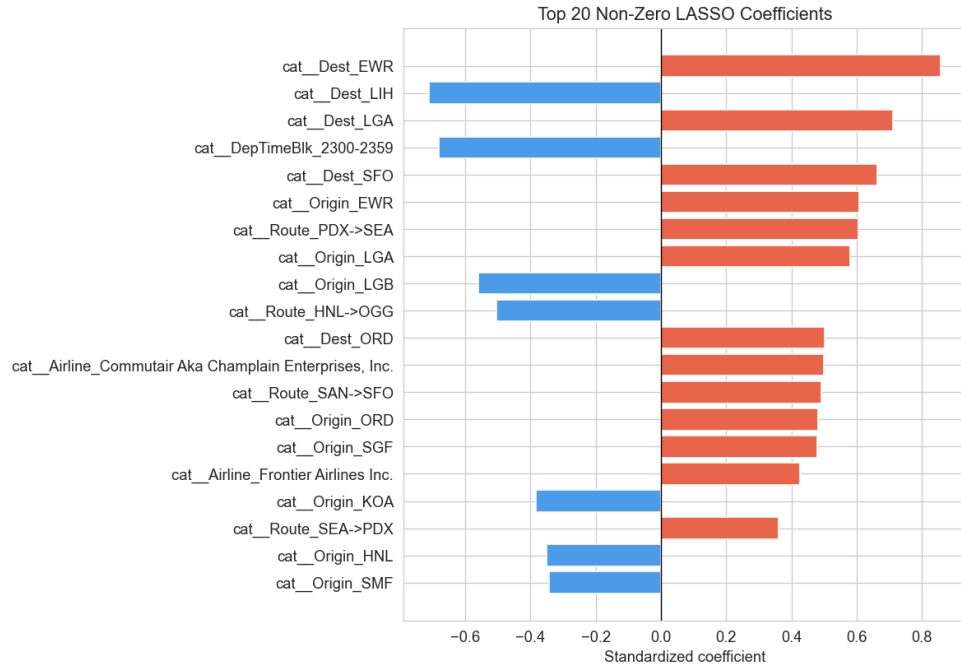


Figure 5. Largest nonzero LASSO coefficients. Positive coefficients increase predicted delay risk, while negative coefficients decrease it.

Table 2. Top random forest feature-group importances.

Feature group	Importance
Departure hour	0.208
Day of month	0.130
Distance	0.124
Scheduled elapsed time	0.110
Month	0.097
Airline	0.096
Destination airport	0.060
Day of week	0.056
Origin airport	0.046
Departure time block	0.040

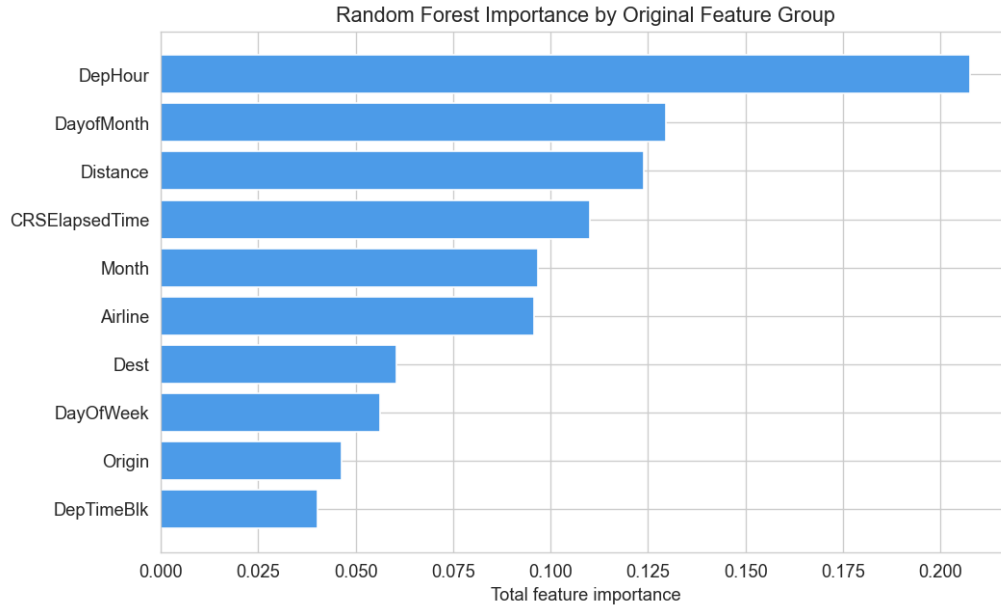


Figure 6. Random forest feature-group importance. Timing variables are highly influential, but carrier, airport, and route information also contribute.

Table 3. Airport cluster summary.

Cluster label	Airports	Delay Rate	Mean Arrival Delay	Flights/Day	Example Airports
Congested Hubs	43	19.4%	14.1	366.8	EW, ORD, LGA, DFW, DEN
Small High-Volatility Airports	67	21.9%	21.1	15.8	ACK, ASE, USA, EGE, HDN
Lower-Risk Diversified Airports	168	16.2%	13.1	28.1	contrast group

5.3 Threshold and Calibration Diagnostics

The validation-selected thresholds remain at 0.5 for the LASSO, random forest, and extra-trees models in the saved threshold analysis. This does not imply that 0.5 is universally optimal for deployment; rather, it means that under the validation objective used in this project, the default threshold provided the best summary tradeoff among the candidate thresholds. The calibration summaries show that predicted probabilities are imperfect, with Brier scores of 0.236 for LASSO, 0.213 for random forest, and 0.224 for extra trees. The random forest therefore has the best combination of ranking and probability quality among the tested models, although its calibration is still not strong enough to treat predicted probabilities as exact risk estimates.

5.4 Airport Clusters

The airport clustering analysis identifies three groups. Table 3 gives the cluster-level summaries.

The clusters are substantively useful. The congested-hub group captures large coordinated systems where delay exposure is high but operational complexity is also high. The small high-volatility group suggests that some lower-volume airports still face elevated delay risk, potentially because thinner schedules and fewer recovery options make disruptions harder to absorb. The lower-risk diversified group provides a contrast class with lower average delay rates and more stable profiles.

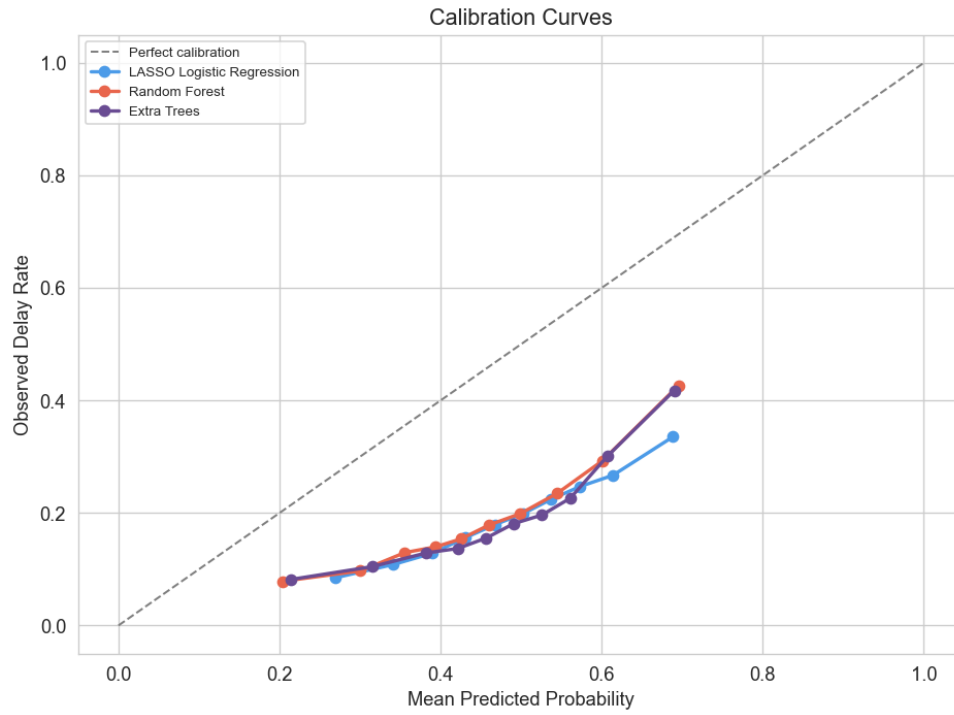


Figure 7. Calibration curves for the predictive models. The models provide useful ranking but imperfect probability calibration.

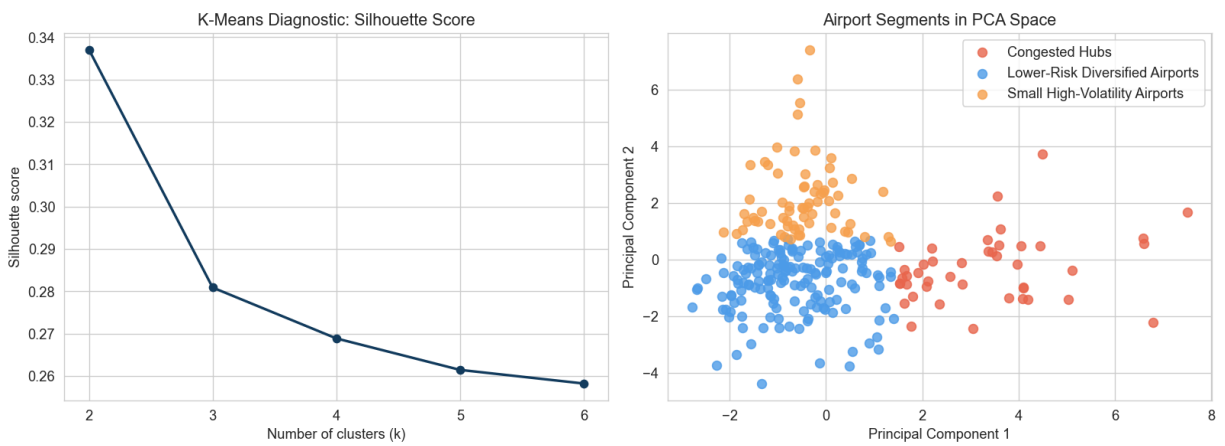


Figure 8. Clustering diagnostics and PCA visualization. Although $k = 2$ has the highest silhouette score, $k = 3$ provides a more interpretable operational segmentation.

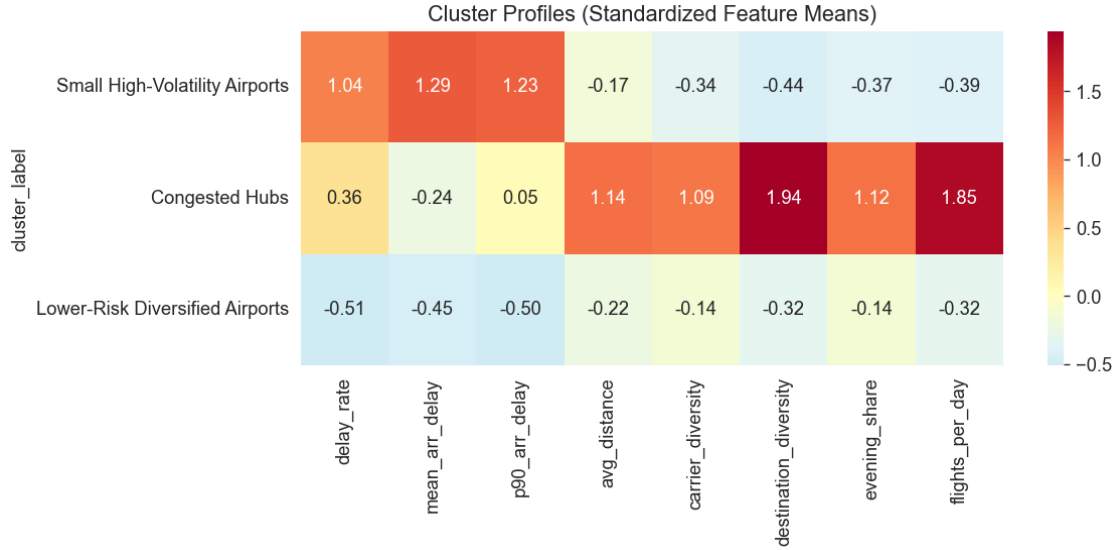


Figure 9. Cluster profile heatmap. The three-cluster solution separates airports by delay risk, traffic intensity, diversity, and operating structure.

6 Discussion

The project produces two main takeaways. First, flight delays are predictable to a useful but limited extent using only pre-departure information. The random forest clearly improves on both the baseline and the LASSO model, while the LASSO model remains valuable for interpretation. Second, airport-level delay environments are heterogeneous in a way that clustering can summarize meaningfully. The airport segmentation helps contextualize the predictive findings by showing that risk is not simply a flight-level phenomenon; it is also embedded in airport operating structure.

The top-decile lift result is especially important from an applied perspective. A model that merely improves overall accuracy would not be very useful, because the delayed class is the minority class. The random forest instead concentrates true delays in the highest-risk decile at more than twice the base rate. This suggests that the model could support use cases such as proactive passenger communication, monitoring higher-risk routes, or prioritizing operational attention. However, the moderate AUC also shows that the model should not be used as a deterministic prediction engine.

The clustering results are also useful because they separate two different forms of risk. Large congested hubs have high traffic volume and operational complexity, while small high-volatility airports have fewer flights but higher average delay severity. This distinction matters because the same intervention may not work for both groups. Hubs may require congestion management and network coordination, while smaller volatile airports may need resilience planning around schedule thinness and recovery options.

7 Limitations

The analysis should not be oversold. The predictive models achieve moderate rather than high discrimination, which is expected given the omitted drivers. Weather shocks, network spillovers, maintenance events, aircraft rotation delays, crew constraints, and real-time congestion are central to delay formation but are not included here. The data also cover only 2019, so the analysis does not claim full temporal generality, especially after major post-2019 changes in air travel demand and operations.

There are also modeling limitations. The 200,000-row modeling cap improves computational feasibility but means the final predictive models are not trained on every cleaned record. The airport clusters use annual summary profiles, which are useful for interpretation but hide within-airport seasonality and day-level

disruptions. Finally, the $k = 3$ cluster solution is chosen partly for interpretability rather than purely for a single diagnostic optimum. This is appropriate for a descriptive segmentation task, but it should be acknowledged openly.

8 Conclusion

This project succeeds at the course's main objective: it poses a substantive data mining question, uses appropriate data, justifies the modeling choices, communicates results clearly, and is honest about uncertainty. The best predictive model, random forest, meaningfully improves delay-risk ranking and identifies a high-risk decile with 2.21 times the baseline delay rate. The airport segmentation adds a complementary unsupervised view by identifying Congested Hubs, Small High-Volatility Airports, and Lower-Risk Diversified Airports.

The final conclusion is practical rather than exaggerated: pre-departure variables contain useful delay-risk information, but they do not fully determine outcomes. A stronger future version would add weather, aircraft rotation, air traffic control, and real-time congestion data, then evaluate whether those additional variables improve both ranking performance and probability calibration.

A Reproducibility Notes

The repository is organized to make the analysis reproducible. The main execution order is:

1. 01_data_prep.ipynb
2. 02_eda.ipynb
3. 03_modeling.ipynb
4. 04_airport_segmentation.ipynb

Alternatively, the full workflow can be run end to end with:

```
python scripts/run_full_pipeline.py
```

The most important generated outputs are stored in `outputs/figures` and `outputs/tables`. The key report artifacts are:

- `model_comparison_metrics.csv` and `model_risk_lift.csv`.
- `threshold_analysis_summary.csv` and `calibration_summary.csv`.
- `airport_cluster_summary.csv`.
- Model-comparison figure, cluster-diagnostics/PCA figure, and calibration-curves figure.

B Statement on AI Use

Generative AI was used as a project support tool rather than as a substitute for analysis. Specifically, AI assistance was used to review the repository for reproducibility, modeling, and presentation gaps; refactor the project into a cleaner pipeline with reusable Python modules; improve notebook structure and project documentation; and suggest clearer narrative framing for the report and presentation. All data cleaning logic, model specifications, clustering choices, figures, tables, and final interpretations remain the responsibility of the project author.

References

- [1] Bureau of Transportation Statistics. *Airline On-Time Performance Data*. U.S. Department of Transportation.

- [2] Kaggle. *Flight Delay Dataset: Combined Flights 2019*. Used as the source distribution for the BTS-derived 2019 domestic flight records.